Beverage Sales Performance Analysis Report

Analysis of sales across products, regions, and categories

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Tool Used: Power Bi

1. Introduction

This report presents an analysis of beverage sales data to evaluate performance across products, customers, and regions using Power BI

2.Objectives.

The goal of this analysis is to better understand how the beverage business is performing and to uncover insights that can support smarter decision-making.

2.1 Analyze Total Revenue and Sales Performance

This analysis focuses on evaluating the overall revenue generated from beverage sales. By looking at total sales performance, we aim to understand how well the business is doing financially and identify any noticeable trends or patterns in revenue over time.

2.2 Identify Top-Performing Products

Another key objective is to determine which products are driving the most sales. By identifying top-performing products, the business can focus on what is working well, while also exploring opportunities to improve or promote lower-performing items.

2.3 Evaluate Sales by Region

This project also looks at how sales differ across various regions. Understanding regional performance helps highlight where the business is strongest and where there may be opportunities for growth or improvement.

2.4 Understand the Impact of Discounts

Discounts can influence customer purchasing behavior, but they can also reduce overall revenue. This analysis aims to assess how discounts are affecting sales and whether they are being used effectively to drive performance without significantly impacting profitability

3. Data Description

The dataset used for this analysis was sourced from Kaggle**,** a well-known platform that provides publicly available datasets for data analysis and machine learning projects. It contains synthetic beverage sales data designed to simulate real-world business transactions.

The dataset includes information on customer purchases, product details, and sales performance. Key columns in the dataset include customer-related fields such as customer ID and region, product-related fields such as product name and category, and sales-related fields such as quantity, unit price, discount, and total price.

Each row in the dataset represents a single sales transaction, capturing what was purchased, by whom, and at what value. This structure makes it possible to analyze sales performance from multiple perspectives, including product performance, customer behavior, and regional trends.

4. Data Cleaning and Transformation

Before building the dashboard, the dataset was carefully reviewed and cleaned to ensure the data was accurate, consistent, and suitable for analysis.

The process began with inspecting the dataset for any inconsistencies such as duplicate records, missing values, and incorrect formatting. Duplicate entries were removed where necessary to ensure that each record represented a valid transaction. A calculated column for Total Revenue was created.

Data types were adjusted to match the nature of each column. For example, numerical fields such as quantity, unit price, discount, and total price were properly formatted as numeric values, while columns such as customer ID and product name were set as text. This ensured that calculations and aggregations would work correctly in Power BI.

To improve the structure of the data, the original dataset was transformed into separate tables. A product table was created by extracting unique product values and removing duplicates. An

index column was then added to generate a unique Product_ID for each product. This allowed for a more structured and efficient way to reference products across the dataset.

Similarly, a customer table was created by selecting relevant customer information and removing duplicate entries to ensure each customer appeared only once. This helped in organizing customer data and supporting accurate analysis by region and customer segments.

The main dataset was then converted into a fact table by retaining only the necessary transactional fields. A merge operation was performed to bring the newly created Product_ID into the fact table by matching product names. This step ensured that relationships could be established between the fact table and the product dimension.

Finally, relationships were created between the tables using common keys such as Customer_ID and Product_ID, enabling a structured data model for analysis.

5.Data Modelling

To ensure efficient analysis and better performance in Power BI, the dataset was transformed into a structured data model using a star schema approach.

The original dataset contained all information in a single table, which can make analysis less efficient and harder to manage. To improve this, the data was split into separate tables based on their roles.

The central table, known as the FactSales table, contains all transactional data such as order ID, customer ID, product ID, quantity, unit price, discount, and total price. This table captures the measurable business events and serves as the core of the model.

Surrounding the fact table are the dimension tables:

- The DimCustomer table stores customer-related information such as customer ID, customer name, and region.
- The DimProduct table contains product details such as product ID, product name, and category.

These dimension tables provide descriptive context that allows the data to be grouped and analyzed from different perspectives.

Relationships were then created between the tables using unique identifiers. The Customer_ID links the DimCustomer table to the FactSales table, while the Product_ID connects the DimProduct table to the FactSales table. These relationships follow a one-to-many structure, where each customer or product can appear multiple times in the sales transactions.

This modeling approach ensures that filters from dimension tables, such as region or product category, correctly affect the sales data. It also improves performance and makes the model easier to understand and maintain.

6.Measures

The DAX measure:

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Total Revenue = SUM (FactSales [Total Price])
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I created and calculated measure in Power BI that adds up all values in the **Total Price** column inside the **FactSales** table.

Simple explanation:

- FactSales = main sales table (fact table)
- Total Price = a column that likely contains the revenue for each individual sale or transaction
- SUM(...) = adds all those values together

7.Dashboard Explanation

7.1 Revenue by Product

This visual breaks down total revenue across individual products, making it easy to see which items are actually driving sales performance.

From this view, we can quickly identify the top-performing products contributing the most to revenue, as well as the lower-performing ones that may not be adding much value. It gives a clear picture of product-level performance across the business.

Business takeaway: This helps the business focus on high-performing products, while also highlighting products that may need pricing adjustments, promotion, or review.

7.2 Revenue by Region

This visual shows how revenue is distributed across different regions, giving a geographic view of sales performance.

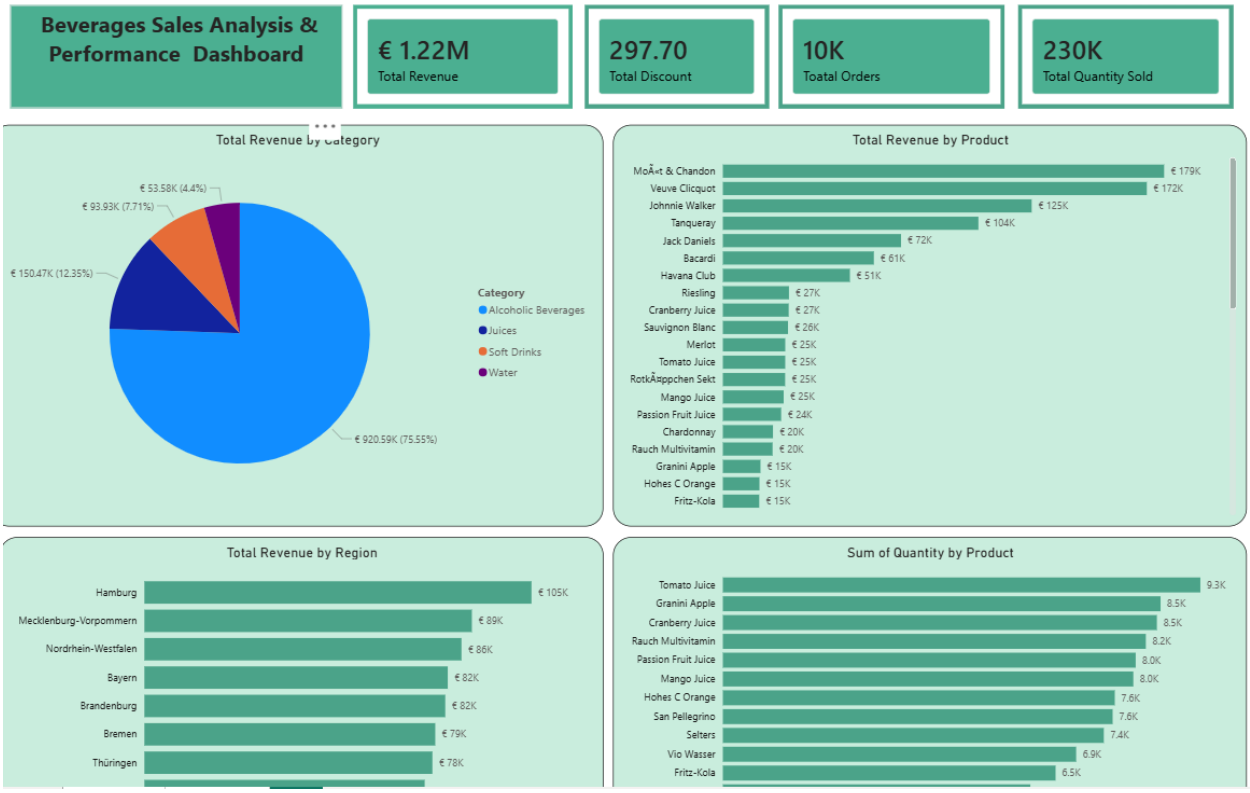
It highlights which regions are generating the strongest revenue and which areas are lagging behind. This makes it easier to understand where the business is performing well and where there may be growth opportunities.

Business takeaway: Strong regions can be further optimized, while weaker regions may require targeted marketing, improved distribution, or customer engagement strategies.

7.3 Category Distribution

This visual presents the distribution of revenue across different product categories, showing how each category contributes to overall sales.

It helps us understand which categories are driving the majority of revenue and whether sales are balanced or concentrated in a few areas.



Business takeaway: This insight supports decisions around inventory planning, product expansion, and category-level strategy to ensure a balanced and profitable product m

8.Key Insights

8.1 Soft Drinks generated the highest revenue
8.2 Water products had the highest sales volume
8.3 The South region recorded the highest revenue
8.4 Discounts reduced overall profit margins

9.Recommendations

Based on the analysis, there are a few practical steps the business can take to improve performance.

First, it would make sense to focus more on the products that are already doing well. These products are clearly driving a large portion of revenue, so increasing their visibility through promotions or ensuring they are always in stock could help boost overall sales even further.

Some products are not performing as strongly. Instead of ignoring them, it would be useful to understand why. It could be due to pricing, lack of awareness, or customer preference. Making small adjustments in these areas could improve their performance.

Looking at regional performance, certain regions are contributing more to total revenue than others. These high-performing regions can be further strengthened, while lower-performing regions may need more attention, such as targeted marketing or better distribution strategies.

Finally, discounts should be used carefully. While they can help increase sales volume, excessive discounting may reduce overall revenue. A more balanced approach to discounts would help maintain profitability while still attracting customers

10. Conclusion

This project provided a clear view of how the beverage business is performing across different products, customers, and regions.

By organizing the data into a structured model and building an interactive dashboard in Power BI, it became easier to identify key trends and patterns in the data. The analysis highlighted which products are driving revenue, how sales vary across regions, and how pricing strategies such as discounts impact overall performance.

The insights gained from this analysis can support better business decisions, especially in areas such as product strategy, regional focus, and pricing. With the right actions based on these findings, the business can improve performance and continue to grow.